FR3 ROS 2 & MoveIt

Practical Lab Guide for Network Setup, FCI, MoveIt, RViz, Controllers, Verification & Recovery

Combined, translated into English, cleaned up, and reorganized from the two supplied lab-note files.

Environment	ROS 2 Humble · franka_ros2 · libfranka · MoveIt 2
Robot	Franka Research 3 (FR3)
Robot / Controller IP	172.16.0.2
Laptop Ethernet IP	172.16.0.1/24
Primary interface	enp3s0
MoveIt planning group	fr3_arm

1. Architecture and Compatibility

libfranka is the low-level library used to communicate with Franka research robots. franka_ros2 provides the ROS 2 integration on top of libfranka. Their versions must therefore be compatible.

> NOTE: Before debugging ROS 2 or MoveIt, first confirm basic network communication and FCI access. A ROS-side problem can otherwise be confused with a network or robot-state problem.

2. Ethernet and FCI Connection

2.1 Direct laptop-to-robot connection through X5, then controller C5

If the Ethernet cable is initially connected directly from the laptop to the robot through port X5, activate FCI from Franka Desk at the robot-side Desk address. Then move the cable to the controller C5 port and configure the laptop interface.

```
sudo nmcli dev set enp3s0 managed no
sudo ip addr flush dev enp3s0
sudo ip route flush dev enp3s0
sudo ip link set enp3s0 up
sudo ip addr add 172.16.0.1/24 dev enp3s0
sudo ip route replace 172.16.0.0/24 dev enp3s0 src 172.16.0.1
sudo ip route flush cache
ping -c 10 172.16.0.2
```

Target result: 0% packet loss.

2.2 Persistent NetworkManager profile for laptop-to-controller C5

When the laptop is connected directly to controller port C5, Franka Desk is available at 172.16.0.2/desk/. A persistent NetworkManager connection can be created as follows:

```
sudo nmcli con delete fr3-fci 2>/dev/null
sudo nmcli con add type ethernet ifname enp3s0 con-name fr3-fci \
  ipv4.method manual \
  ipv4.addresses 172.16.0.1/24 \
  ipv6.method ignore
sudo nmcli con up fr3-fci

ip addr show enp3s0
ip route get 172.16.0.2
ping -c 10 172.16.0.2
```

A known successful test in the notes showed 10/10 replies and 0% packet loss.

2.3 Confirm robot state with libfranka

After Desk is ready, unlock the robot as required and activate FCI. Then confirm that robot state can be read:

```
/home/danai/franka_ros2_ws/install/libfranka/bin/echo_robot_state 172.16.0.2
```

An alternative path used in the notes is:

```
/home/danai/franka_ros2_ws/build/libfranka/examples/echo_robot_state 172.16.0.2
```

3. Workspace Build and ROS 2 Environment

For a clean rebuild of the FR3 ROS 2 workspace:

```
cd ~/franka_ros2_ws
source /opt/ros/humble/setup.bash

rm -rf build install log

colcon build --symlink-install \
```

```
  --event-handlers console_direct+ \
  --cmake-args \
  -DCMAKE_BUILD_TYPE=Release \
  -DBUILD_TESTS=OFF \
  -DBUILD_TESTING=OFF \
  -DBUILD_EXAMPLES=ON

source ~/franka_ros2_ws/install/setup.bash
ros2 pkg list | grep franka
```

NOTE: In every new terminal, source ROS 2 Humble first and then the FR3 workspace setup file.

4. MoveIt Workflow - Fake Hardware / Simulation

Terminal 1 - Launch MoveIt with fake hardware

```
source /opt/ros/humble/setup.bash
source ~/franka_ros2_ws/install/setup.bash

ros2 launch franka_fr3_moveit_config moveit.launch.py \
  robot_ip:=172.16.0.2 \
  arm_id:=fr3 \
  use_fake_hardware:=true
```

Leave this terminal running.

Terminal 2 - Run the semantic relay

```
source /opt/ros/humble/setup.bash
source ~/franka_ros2_ws/install/setup.bash
python3 ~/fr3_moveit_tools/semantic_relay.py
```

Expected messages include:

```
Loaded robot_description from /move_group.
Loaded robot_description_semantic from /move_group.
```

Terminal 3 - Open RViz again if needed

If the RViz instance opened by the launch file does not show the planning group correctly:

```
source /opt/ros/humble/setup.bash
source ~/franka_ros2_ws/install/setup.bash

rviz2 -d ~/franka_ros2_ws/install/franka_fr3_moveit_config/share/franka_fr3_moveit_config/rviz/moveit.rviz
```

In RViz, go to MotionPlanning → Planning Request → Planning Group and select fr3_arm. Then use Plan and Execute. With fake hardware, the model should move in RViz.

5. MoveIt Workflow - Real FR3

SAFETY: Before any real movement, keep the workspace clear, keep the emergency stop accessible, and verify the robot state in Franka Desk. For the first motion, use only a small change and visually inspect the planned trajectory before execution.

5.1 Desk checklist before starting

- System: Ready
- FCI: ON
- Arm connected
- Joints unlocked
- No self-collision
- No reflex/error
- Emergency stop within immediate reach

5.2 Terminal 1 - Launch MoveIt for the real robot

```
source /opt/ros/humble/setup.bash
source ~/franka_ros2_ws/install/setup.bash

ros2 launch franka_fr3_moveit_config moveit.launch.py \
  robot_ip:=172.16.0.2 \
  arm_id:=fr3 \
  use_fake_hardware:=false
```

Leave this terminal running.

5.3 Terminal 2 - Run the semantic relay

```
source /opt/ros/humble/setup.bash
source ~/franka_ros2_ws/install/setup.bash
python3 ~/fr3_moveit_tools/semantic_relay.py
```

Confirm that robot_description_semantic is loaded.

5.4 Terminal 3 - RViz

```
source /opt/ros/humble/setup.bash
source ~/franka_ros2_ws/install/setup.bash

rviz2 -d ~/franka_ros2_ws/install/franka_fr3_moveit_config/share/franka_fr3_moveit_config/rviz/moveit.rviz
```

In RViz: MotionPlanning → Planning Request → Planning Group → fr3_arm.

- Make a small change with the interactive marker.
- Press Plan.
- Visually inspect the complete trajectory.
- Press Execute only if the planned motion is safe.

6. Verify MoveIt and Semantic Description

While MoveIt is active, check whether the semantic robot description is available:

```
ros2 param get /move_group robot_description_semantic
```

A large SRDF text response indicates that /move_group has the semantic description.

With the semantic relay running:

```
ros2 topic list | grep robot_description
```

Expected topics:

```
/robot_description
/robot_description_semantic
```

Check that the semantic description is actually published:

```
ros2 topic echo /robot_description_semantic --once
```

If SRDF content containing groups is printed, RViz should be able to expose the MoveIt planning group.

7. Safe State-Only Bringup

To connect to the real FR3 and activate broadcasters without commanding motion:

```
source /opt/ros/humble/setup.bash
source ~/franka_ros2_ws/install/setup.bash

ros2 launch franka_bringup franka.launch.py \
  robot_ip:=172.16.0.2 \
  robot_type:=fr3 \
  load_gripper:=false \
  use_fake_hardware:=false
```

> NOTE: This bringup does not intentionally command robot motion. It connects the FR3 and enables the relevant broadcasters.

8. Controller Examples

8.1 Joint position example

```
ros2 launch franka_bringup example.launch.py \
  robot_ips:=172.16.0.2 \
  controller_names:=joint_position_example_controller
```

Produces a simple repeated joint-space motion.

8.2 Model example

```
ros2 launch franka_bringup example.launch.py \
  robot_ips:=172.16.0.2 \
  controller_names:=model_example_controller
```

Useful for checking model-related quantities and generally a good low-motion validation step.

8.3 Cartesian pose example

```
ros2 launch franka_bringup example.launch.py \
  robot_ips:=172.16.0.2 \
  controller_names:=cartesian_pose_example_controller
```

Can command the end effector along a Cartesian trajectory. Use only with a clear workspace.

8.4 Cartesian orientation example

```
ros2 launch franka_bringup example.launch.py \
  robot_ips:=172.16.0.2 \
  controller_names:=cartesian_orientation_example_controller
```

Primarily changes end-effector orientation. Pay particular attention to wrist and gripper clearance.

8.5 Joint impedance example

```
ros2 launch franka_bringup example.launch.py \
  robot_ips:=172.16.0.2 \
  controller_names:=joint_impedance_example_controller
```

8.6 Joint impedance with IK

```
ros2 launch franka_bringup joint_impedance_with_ik_example_controller.launch.py \
```

```
  robot_ips:=172.16.0.2
```

SAFETY: The original notes recommend leaving impedance-with-IK for later, after understanding the expected motion and ensuring a completely clear workspace.

9. Controllers Requiring Extra Caution

The notes explicitly flag the following velocity controllers as more dynamic and not suitable as first tests:

```
ros2 launch franka_bringup example.launch.py \
  robot_ips:=172.16.0.2 \
  controller_names:=joint_velocity_example_controller

ros2 launch franka_bringup example.launch.py \
  robot_ips:=172.16.0.2 \
  controller_names:=cartesian_velocity_example_controller
```

SAFETY: Do not test joint_velocity_example_controller, cartesian_velocity_example_controller, or mobile_cartesian_velocity_example_controller near people or obstacles.

10. Recommended Test Order

Step	Controller	Purpose / note
1	joint_position_example_controller	Already tested in the notes
2	model_example_controller	Model / validation step
3	cartesian_pose_example_controller	Cartesian position motion
4	cartesian_orientation_example_controller	Orientation motion
5	joint_impedance_example_controller	Higher caution
6	joint_impedance_with_ik_example_controller	Run only after understanding expected motion

11. Monitoring During Every Test

Use a second terminal to monitor the active ROS 2 controllers and joint state:

```
ros2 control list_controllers
ros2 topic echo /joint_states --once
ros2 control list_controllers | grep active
```

12. Safety Rule Before Switching Controllers

- Stop the current controller or launch process with Ctrl+C.
- Wait until the robot has fully stopped.
- Check Franka Desk: System Ready, FCI ON, no self-collision, and no reflex/error.
- Only then start the next controller.

NOTE: If the robot enters an error state, the notes recommend using Franka Desk in Prog mode to recover and manually place the robot in a safe pose before continuing.

13. RViz Recovery

If RViz crashes, close only RViz and reopen it. Keep the move_group and semantic relay terminals running.

```
pkill -9 -f rviz2

rviz2 -d ~/franka_ros2_ws/install/franka_fr3_moveit_config/share/franka_fr3_moveit_config/rviz/moveit.rviz
```

14. Joint-Trajectory Experiment Command

The supplied notes include the following command for the experiment trajectory script:

```
python3 ~/Desktop/fr3_experiments/scripts/send_exp1_joint_trajectory.py \
  --joint-topic /joint_states \
  --action-name /fr3_arm_controller/follow_joint_trajectory \
  --duration 8.0 \
  --delta "0.20,0,0,0,0,0,0"
```

15. Compact Lab Checklist

Phase	Check
Network	Laptop 172.16.0.1/24 → robot/controller 172.16.0.2; ping has 0% packet loss
Desk	System Ready; FCI ON; arm connected; joints unlocked; no reflex/error
libfranka	echo_robot_state returns robot state
Terminal setup	Source /opt/ros/humble/setup.bash, then workspace install/setup.bash
MoveIt	Launch fake or real hardware mode as appropriate
Semantic relay	robot_description and robot_description_semantic loaded
RViz	Planning group = fr3_arm
Before Execute	Small first motion; inspect planned trajectory; clear workspace
Monitoring	list_controllers + joint_states
Switching tests	Ctrl+C → full stop → Desk check → next controller

Prepared from the supplied notes only. Commands and environment-specific paths were retained from those notes; duplicated material was consolidated and Greek explanatory text was translated into English.